Math 420: Project (Version 6) Elise Murphy

i- $\pi A = 0$, $\sum_{i=0}^{27} P_i = 1$

$\lambda P_i = \mu P_{i+1} \Rightarrow P_{i+1} = \frac{\lambda}{\mu} P_i$ let $\rho = \frac{\lambda}{\mu} = \frac{4.9}{7.3}$

$P_i = P_0 \rho^i$ for $i = 0, \ldots, 27$

Normalize : $P_0 \sum_{i=0}^{27} \rho^i = 1$ $P_0 = \frac{1-\rho}{1-\rho^{28}}$ $P_i = \frac{1-\rho}{1-\rho^{28}} \rho^i$, $i = 0, 1, \ldots, 27$

ii- $P(X_t = 0) = P_0$

$P_0 = \frac{1-\rho}{1-\rho^{28}}$ $\rho = \frac{\lambda}{\mu}$

The mechanic doesn't have to work 32.88 % of the time.

iii- $E[X_t] = \sum_{i=0}^{27} i P_i$; know $P_i = P_0 \rho^i$ $\rho = \frac{4.9}{7.3}$ $P_0 = \frac{1-\rho}{1-\rho^{28}}$

$E[X_t] = \sum_{i=0}^{27} i P_0 \rho^i$

iv- $P(X_t \geq 5) = \sum_{i=5}^{27} P_i$

$P(X_t \geq 5) = P_0 \cdot \frac{\rho^5 (1-\rho^{23})}{1-\rho}$

v- $\mu_i = \begin{cases} 0, & i = 0 \\ \mu, & i = 1 \\ 2\mu, & i \geq 2 \end{cases}$, $\lambda = 4.9$

For $\tilde{A}$ has: $i \to i+1$ w/ rate λ

$i \to i-1$ w/ rate μ_i

$\tilde{A}_{i,i+1} = \lambda$

$\tilde{A}_{i,i-1} = \begin{cases} \mu, & i = 1 \\ 2\mu, & i = 2, \ldots, 27 \end{cases}$

$\tilde{A}_{i,i} = -$ (total rate of state i)

$$\tilde{A} = \begin{bmatrix} -\lambda & \lambda & 0 & \cdots & 0 \\ \mu & -(\lambda+\mu) & \lambda & \cdots & 0 \\ 0 & 2\mu & -(\lambda+2\mu) & \lambda & \cdots \\ \vdots & \vdots & \ddots & \ddots & \ddots \\ 0 & \cdots & 0 & 2\mu & -2\mu \end{bmatrix}$$

$\tilde{P}_i = \tilde{P}_0 \frac{\lambda^i}{\mu_1 \mu_2 \dots \mu_i}$

$\tilde{P}_1 = \tilde{P}_0 \frac{\lambda}{\mu}$

for $i \geq 2$, $\tilde{P}_i = \tilde{P}_0 \frac{\lambda^i}{\mu (2\mu)^{i-1}}$

$\tilde{P}_i = \tilde{P}_0 \left(\frac{1}{\mu}\right)\left(\frac{\lambda}{2\mu}\right)^{i-1}$, $i \geq 1$

vi- $E[\tilde{X}] = \sum_{i=0}^{27} i \tilde{P}_i$ → # of expected forklifts = 0.74 forklifts

Prob of first mechanic is busy

$P(\text{first busy}) = P(\tilde{X} \geq 1) = 1 - \tilde{P}_0 \approx 50.24\%$

Prob of mechanic #2 being called in :

$P(\text{second called}) = \sum_{i=2}^{27} \tilde{P}_i = 1 - \tilde{P}_0 - \tilde{P}_1 \approx 14.87\%$

Prob of 5+ forklifts in repair facility.

$P(\tilde{X}_t \geq 5) = \sum_{i=5}^{27} \tilde{P}_i \approx 0.64\%$

iiv- Adding the second mechanic improves the efficency of the repair facility. The expected # of forklifts in repair decreases from 2.04 to 0.74, the probability of having 5 or more forklifts drops from 13.42% to 0.64%. And the first mechanic is less busy than before, with the second mechanic only needed 14.87% of the time.

viii- Second mechanic : $290 a day

When 2+ forklifts, lease replacements : $25 /day per forklift.

Second mechanic : $ 42.17

Lease replacement forklifts : $ 31.25

More cost-effective to lease replacement forklifts until the first mechanic can catch up on repairs.

ix- When the mechanic's pay is less than $203.07, hiring a second mechanic is most cost-effective. But as long as the 2nd mechanic's pay is above that threshold leasing replacement forklifts is most cost-effective.